Dalve Pavankumar Ravindra

me24btech11019@iith.ac.in | pkumardalve@gmail.com | +91 8180974182 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.91
Parimal Vidyalaya, latur	Class XII (HSC)	2023	77.83%
Shri Keshavraj Vidyalaya, Latur	Class X (SSC)	2021	98.80%

SKILLS

Programming: C, C++, Python

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems.

Machine Learning / Data: TensorFlow, Convolutional Neural Networks (CNN), NumPy, Matplotlib, LangChain, LLMs (Gemini API)

Developer Tools: Git, GitHub, MATLAB, VS Code, Streamlit

Design & Modeling Tools: Solid Edge, Fusion 360

PROJECTS

LangChain Chatbot with Gemini (LLM-Powered Conversational App):-

- Built a Streamlit-based conversational AI web application integrating Google's Gemini API through LangChain for real-time, streaming chat responses.
- Designed a modular prompt-chaining architecture using LangChain's ChatPromptTemplate and output parsers to structure and control LLM behavior.
- Implemented session-based chat history and token-by-token response streaming, along with configurable model selection and secure API key handling for a seamless user experience.

RankMySketch (CNN-Powered Multiplayer Drawing Game):-

- Developed a backend processing pipeline and trained a CNN using TensorFlow to classify real-time sketches against target prompts.
- Built a scoring system that computed percentage-based visual accuracy, quantifying how closely each sketch matched the intended prompt.
- Automated dynamic player ranking based on live scores, enabling a competitive multiplayer drawing experience.

Interactive Physics Simulation GUI (Python):-

- Built a desktop application in Tkinter to provide an intuitive GUI for simulating complex kinematic systems, including four-bar linkages and slider-crank mechanisms.
- Used Matplotlib to render real-time, dynamic animations, allowing users to visualize mechanism motion and joint behavior as parameters were adjusted.
- Designed the interface to let users configure link lengths and initial conditions, enabling interactive exploration of mechanical motion in real time.

POSITIONS OF RESPONSIBILITY

General Secretary - Mixed Martial Arts Club IITH

- **Kickboxing Tournament Lead (Diesta):** Spearheaded end-to-end logistics, safety protocols, and participant scheduling for a campus-wide kickboxing tournament; executed an injury-free event that significantly increased club visibility and campus engagement.

Competition Co-Ordinator - Elan & nVision Club

- **Competition Management:** Orchestrated 3 major festival events (Battle of the Bands, Short Film, BGMI Esports); designed standardised rulebooks and structured point-calculation systems for dynamic esports leaderboards, ensuring error-free evaluations across 100+ participating teams.

Informals Deputy Co-Ordinator - Elan & nVision Club

- **On-Ground Event Engagement:** Managed scheduling, resource allocation, and crowd logistics for 5+ interactive recreational events (including board game tournaments), engaging 200+ attendees and ensuring smooth on-ground operations throughout the festival.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures & Applications

Mathematics: Differential Equations, Probability, Statistics, Calculus

EXTRACURRICULAR

Music: Guitar

Sports: Cricket, MMA